Global AI Adoption & Market Landscape

2019 – 2025 | End-to-End Data Analytics Project

Python	MySQL	Power BI	pandas	SQLAlchemy

6	14	8	517	5
Datasets	Countries	Industries	Raw Rows	Dashboard Pages

Prepared by	Data Analytics Portfolio Project
Analysis Period	Industry: 2019-2025 | Geographic & Cloud: 2019-2024
Tools	Python | pandas | SQLAlchemy | MySQL | Power BI
Scope	14 Countries | 8 Industries | 4 Cloud Providers | 517 Raw Rows
Dashboard	5-page interactive Power BI | 20+ DAX measures

1. Executive Summary

This report presents the findings of a comprehensive data analytics project examining global AI adoption trends from 2019 to 2025. Six independent datasets — spanning 14 countries, 8 industries, and 4 cloud providers — were cleaned, merged, and analyzed using Python, stored and queried in MySQL, and visualized in a 5-page interactive Power BI dashboard.

The analysis surfaces three business-critical findings: AI adoption has doubled globally in six years with significant industry-level disparity; seven high-readiness countries remain under-monetized, representing the clearest market opportunity in the dataset; and Microsoft Azure is outpacing AWS in growth rate, driven visibly by the 2023 OpenAI investment.

73.4%	$715.7B	52.2%	45.1%	$401B
Global AI Adoption 2025	Total Market Revenue 2019–24	Revenue CAGR 2019–2024	Avg Pilot-to- Production Rate	Projected 2025 Revenue

2. Business Problem

Organizations and investors making AI strategy decisions face a fragmented information landscape. Adoption data, investment flows, policy readiness, enterprise spending patterns, and cloud infrastructure metrics all exist — but are published separately by different research organizations with no unified view.

Three core questions drive this project:

Q1	Which industries and regions are leading AI adoption — and which are lagging behind?
Q2	Where does high AI readiness meet low market revenue, creating an investment opportunity gap?
Q3	How are AWS, Azure and Google Cloud competing for cloud AI dominance, and who is gaining share?

3. Data Sources

Six publicly referenced datasets were sourced, each covering a distinct analytical dimension. Together they form a 517-row cross-sectional view of the global AI landscape from 2019 to 2025.

Dataset	Source	Rows	Coverage
mckinsey_ai_survey.csv	McKinsey State of AI	118	Industry adoption rates, AI function, reg
stanford_ai_index.csv	Stanford HAI Index	84	Country investment, patents, research output
oecd_ai_policy.csv	OECD AI Observatory	108	Policy count, readiness score, talent score
statista_ai_market.csv	Statista	98	Market revenue, CAGR by country

idc_ai_spending.csv	IDC	85	Enterprise spending, use cases, barriers
cloud_provider_revenue.csv	Amazon / Alphabet Annual Reports	24	AWS, Azure, GCP revenue and market shares

Industry data covers 2019–2025. Geographic and cloud data covers 2019–2024. All figures are reported in USD billions unless otherwise stated.

4. Methodology

4.1 Data Pipeline

The end-to-end pipeline was built in Python using pandas for transformation and SQLAlchemy for automated MySQL upload. The pipeline executes nine sequential steps:

Step 1: Connect to MySQL and create the ai_adoption_db database

Step 2: Load all 6 raw CSVs into pandas DataFrames

Step 3: Standardise column names (lowercase, strip whitespace) and remove exact duplicates

Step 4: Clean each dataset — type casting, null imputation with median, text normalisation, country name mapping

Step 5: Exploratory Data Analysis — summary statistics, adoption trends, outlier detection via IQR method

Step 6: Merge into 3 final analytical tables (ai_industry, ai_geo, ai_cloud)

Step 7: Export 3 clean CSVs for Power BI import

Step 8: Upload all 3 tables to MySQL via SQLAlchemy to_sql() — auto-creates tables, no manual DDL needed

Step 9: Verify row counts from MySQL to confirm successful load

Key Design Decision — Preventing Duplicate Columns on Merge

Multiple source datasets shared column names (region, country_code). A naive merge would create region_x / region_y suffix columns silently, corrupting all downstream calculations. Solution: overlapping columns were explicitly dropped from secondary sources before merging. Result: zero _x/_y suffix columns across all 3 final tables.

4.2 Output Tables

Table	Source Datasets	Rows	Columns	Year Range
ai_industry	McKinsey + IDC	238	11	2019–2025
ai_geo	Stanford + OECD + Statista	84	18	2019–2024
ai_cloud	Cloud providers (standalone)	24	8	2019–2024

4.3 SQL Analysis

Nine analytical queries were executed in MySQL Workbench against ai_adoption_db, covering:

- Global adoption trend by year — aggregation with GROUP BY, AVG, MIN, MAX
- Regional AI spending totals — SUM and ORDER BY ranking

- Country investment rankings — cumulative SUM across years
- Live cloud market share — window function: SUM() OVER (PARTITION BY year)
- 5-year CAGR per country — POWER() with NULLIF() for safe division
- Industry scaling tiers — CASE WHEN classification on pilot-to-production rate
- Use case adoption comparison — NLP vs Computer Vision vs Automation vs Predictive Analytics
- G20 vs Non-G20 investment gap — CASE-based grouping and comparative aggregation
- Readiness score improvement 2019 to 2024 — conditional MIN/MAX CASE WHEN pivot

4.4 Power BI Dashboard

The 3 clean CSVs were loaded into Power BI Desktop. Over 20 DAX measures were written covering deduplication-safe averages, dynamic KPI cards, CAGR-based revenue projection, and opportunity scoring. All 5 pages include cross-filtering slicers for year, region, income group, and category.

DAX Challenge — Deduplication of Join Artifacts

The left join between McKinsey and IDC data created duplicate rows per industry-year-region combination. A plain AVERAGE() function returned inflated adoption rates. Solution: AVERAGEX(SUMMARIZE(...)) was used to collapse duplicates before averaging — a common real-world pattern when working with multi-source joined data in Power BI.

5. Key Findings

5.1 Global AI Adoption Trend

Global AI adoption more than doubled from 33.9% in 2019 to 73.4% in 2025, with all regions accelerating sharply after 2021. North America leads throughout, followed closely by Asia-Pacific. Developing Markets entered the tracked dataset only from 2023, reflecting IDC's expanding coverage rather than a gap in actual activity.

Year	Global Avg Adoption	North America	Europe	Asia-Pacific
2019	33.9%	33.5%	33.4%	36.0%
2020	38.9%	39.1%	38.2%	41.3%
2021	44.2%	44.1%	42.6%	46.0%
2022	48.7%	48.6%	47.0%	50.7%
2023	52.7%	52.9%	51.0%	54.3%
2024	61.6%	62.8%	59.2%	61.0%
2025	73.4%	73.9%	69.2%	70.7%

5.2 Industry Adoption Analysis

Significant adoption disparity exists across the 8 industries tracked. Technology leads with a 68% average, while Healthcare sits at 35%, the lowest of all sectors. The gap between the highest and lowest adopting industries is 33 percentage points, reflecting structural differences in regulatory environment, data availability, and talent access.

Industry	Avg Adoption	Scaling Tier	Primary Barrier
Technology	68%	Scale Leader	Talent Gap
Financial Services	58%	Scale Leader	Data Quality
Professional Services	58%	Scale Leader	Talent Gap
Media & Telecom	52%	Moderate Scaler	Talent Gap
Retail & Consumer Goods	48%	Moderate Scaler	Integration Complexity
Manufacturing	37%	Slow Scaler	Legacy Systems
Energy & Materials	36%	Slow Scaler	Data Quality
Healthcare	35%	Slow Scaler	Regulatory Compliance

Talent Gap is the most prevalent barrier, appearing across 43 records. Regulatory Compliance, while less frequent overall, is concentrated in healthcare and is the primary reason the sector lags despite high digitization levels in clinical data.

5.3 Geographic Investment & Readiness

The United States accounts for $326.8B of cumulative AI investment from 2019 to 2024 — more than five times China's $64.4B. North America holds 55.5% of total global AI market revenue. Despite these figures, several countries show strong AI readiness scores with significantly underperforming market revenue, pointing to untapped commercial opportunities.

Country	Avg Readiness Score	Total Investment ($B)	Market Revenue ($B)	Quadrant
United States	91.7	326.8	379.9	Leader
United Kingdom	85.8	19.8	43.1	Leader
Japan	84.3	7.4	33.4	Leader
Singapore	88.1	6.0	11.6	Opportunity
Canada	85.1	11.2	17.5	Opportunity
Germany	83.4	10.2	28.5	Opportunity
Australia	83.3	4.2	8.7	Opportunity
South Korea	82.6	6.4	11.5	Opportunity
France	81.3	8.0	17.8	Opportunity
Israel	81.0	16.3	15.2	Opportunity
China	78.4	64.4	116.7	High Revenue
India	60.0	9.5	17.1	Emerging
UAE	75.4	4.8	7.7	Emerging
Brazil	55.4	3.5	7.0	Emerging

Opportunity Gap Finding

Seven countries — Singapore, Australia, Japan, Canada, South Korea, Germany, and France — score above 80 in AI readiness but show disproportionately low market revenue relative to that score. These markets have strong digital infrastructure, policy frameworks, and AI talent pipelines but remain under-served by AI vendors. This is the clearest commercial opportunity signal identified in the dataset.

India recorded the highest readiness score improvement of any country — +15.8 points from 2019 to 2024 — despite being classified as a Lower Middle-Income economy. Brazil (+13.7) and UAE (+13.7) follow. Emerging markets are closing the readiness gap faster than absolute investment figures suggest.

5.4 Cloud Platform Competition

AWS, Microsoft Azure, and Google Cloud collectively held 67% of the global cloud AI market in 2024, with the remaining 33% distributed across other providers. The competitive dynamics have shifted significantly since 2019 — AWS is losing share while Azure is gaining it at pace.

Provider	Revenue 2019	Revenue 2024	Share 2019	Share 2024	Avg YoY Growth	5-Yr CAGR
AWS	$25.7B	$107.6B	33%	30%	27.4%	33.1%
Microsoft Azure	$13.8B	$71.8B	18%	25%	44.3%	39.1%
Google Cloud	$8.9B	$43.2B	9%	12%	39.6%	37.1%
Others	$25.7B	$63.5B	40%	33%	19.9%	19.8%

Azure Growth Inflection — OpenAI Effect

Azure's year-on-year growth showed a visible acceleration in 2023 and 2024, coinciding with Microsoft's $10B investment in OpenAI and the subsequent integration of ChatGPT workloads into Azure infrastructure. This is directly observable in the YoY growth data — Azure jumped from 24.7% growth in 2023 to 40.8% in 2024 — the only provider to accelerate in the final year while others showed growth maturation or decline.

6. Power BI Dashboard Overview

The findings above are presented in a 5-page interactive Power BI dashboard built on the 3 clean output tables. All pages include cross-filtering slicers enabling dynamic exploration by year, region, income group, industry, and adoption tier.

#	Dashboard Page	Primary Insight
1	Executive Overview	73.4% global AI adoption in 2025 | $715.7B total revenue | 45% pilot-to-production rate
2	Geographic Analysis	USA investment 5x China | N. America 55.5% revenue share | 7 opportunity-gap countries
3	Industry Adoption Analysis	Technology leads 68% | Healthcare lags 35% | Talent Gap is #1 barrier across all industries
4	AI Platforms & Market Share	Azure fastest growing (44.3% avg YoY) | AWS losing share 33% to 30% | OpenAI effect visible
5	Future Opportunity & Gap	$401B projected 2025 | India readiness +15.8 pts | 7 high-readiness under-monetized markets

7. Conclusions & Recommendations

1. AI adoption has doubled — but scaling has not.

Global adoption rose from 33.9% to 73.4% in six years. However, only 45.1% of AI pilots reach production. The majority of enterprise AI projects remain stuck at pilot stage, indicating that the challenge has shifted from adoption intent to operational scaling.

2. Industry-level strategy must differ.

A 33-percentage-point gap between Technology and Healthcare reflects fundamentally different operating environments. Vendors targeting Healthcare must address regulatory compliance first — not talent or data pipelines — as it is the dominant barrier in that sector specifically.

3. Seven markets represent the most immediate commercial opportunity.

Singapore, Australia, Japan, Canada, South Korea, Germany, and France all score above 80 in AI readiness yet remain significantly under-served in market revenue. These markets have the infrastructure, policy frameworks, and talent to absorb AI solutions — they simply have not been prioritised proportionally by vendors relative to their US and Chinese counterparts.

4. Azure is the platform to watch.

AWS dominates in absolute revenue but is on a downward share trajectory. Azure's growth rate is the highest of any major provider and accelerating — the only major provider to grow faster in 2024 than in 2023. Organizations building multi-year AI infrastructure commitments should factor in Azure's momentum alongside AWS's current scale advantage.

5. Emerging markets are closing faster than investment data suggests.

India improved AI readiness by 15.8 points in five years despite Limited investment relative to G20 peers. This pace of improvement — driven by government policy, talent development, and startup activity — suggests emerging markets will reach commercial viability earlier than traditional investment-based forecasts would indicate.

8. Project Files

All project files are available in the GitHub repository:

File / Folder	Description
Python code/	AI_Adoption_Analysis.py — full pipeline: cleaning, EDA, merge, MySQL upload
Raw data from all sources/	6 original source CSVs: McKinsey, Stanford, OECD, Statista, IDC, Cloud
Cleaned files (Python Output)/	3 merged output CSVs: ai_industry, ai_geo, ai_cloud
SQL-Queries/	9 analytical SQL queries + DAX measures for Power BI

Dashboard-snaps/	6 dashboard page screenshots (all 5 pages + index)
Power BI Dashboard/	AI Adoption Dashboard.pbix — 5-page interactive dashboard
PPT Presentation/	AI_Adoption_Analysis.pptx — 12-slide project presentation
README.md	Project overview with dashboard screenshots and key findings

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Data Sources: McKinsey State of AI · Stanford HAI Index · OECD AI Observatory · Statista · IDC · Amazon / Alphabet Annual Reports

Analysis period: 2019–2025 (Industry) · 2019–2024 (Geographic & Cloud)